React useState Hook Tutorial

Introduction

The `useState` hook is a fundamental part of React that allows you to add state to functional components. Before hooks, state was only available in class components. With `useState`, you can easily manage and update state in your functional components.

Basic Usage

Here's how you use the `useState` hook: (playcode.io/react)

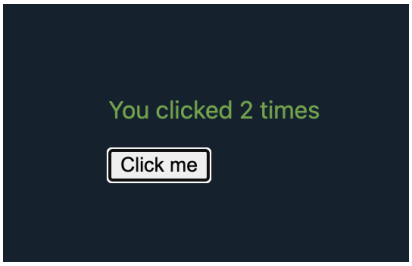
```
import React, { useState } from 'react';

function Counter() {
  // Declare a state variable named 'count', initialized to 0
  const [count, setCount] = useState(0);

  return (
    <div>
      <p>You clicked {count} times</p>
      <button onClick={() => setCount(count + 1)}>
        Click me
      </button>
    </div>
  );
}


export function App(props) {
  return (
    <div className='App'>
    <Counter />
    </div>
  );
}
```

Explanation: - `useState(0)` initializes the state variable `count` to 0. - `setCount` is a function that updates the value of `count`. - When the button is clicked, `setCount(count + 1)` increases the count by 1.



Storing Other Types of State

You can use `useState` to store strings, arrays, objects, and more.

Example: Storing an object

```
import React, { useState } from 'react';

function UserProfile() {
  const [user, setUser] = useState({ name: 'Alice', age: 25 });

  return (
    <div>
      <p>Name: {user.name}</p>
      <p>Age: {user.age}</p>
      <button onClick={() => setUser({ ...user, age: user.age + 1 })}>
        Increase Age
      </button>
    </div>
  );
}
export function App(props) {
  return (
    <div className='App'>
    <UserProfile />
    </div>
  );
}
```

Explanation: - The state variable `user` is an object. - To update the age, we use the spread operator (`...user`) to keep the other properties unchanged.



Exercise

Task: Create a simple React component called `NameForm` that: - Uses `useState` to store a user's name. - Renders an input field and a button. - When the button is clicked, displays a greeting with the entered name.

Example Output:

```
Hello, John!
```

Starter Code:

```jsx
import React, { useState } from 'react';

function NameForm() {
  // Your code here

  return (
    <div>
      {/* Your code here */}
    </div>
  );
}

export default NameForm;
```

Summary

- `useState` lets you add state to functional components.
- You can store any type of value in state.
- Always use the setter function to update state.

For more information, see the React documentation on useState.